Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Digital signal and Image process (01CT1513)	**Aim: Simulate linear convolution and circular convolution on discrete time signals.**	
Experiment - 2	**Date: 18/07/2026**	**Enrolment No:92400133125**

Aim : Simulate linear convolution and circular convolution on discrete time signals.

Github : https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing/tree/main/Experiment%202

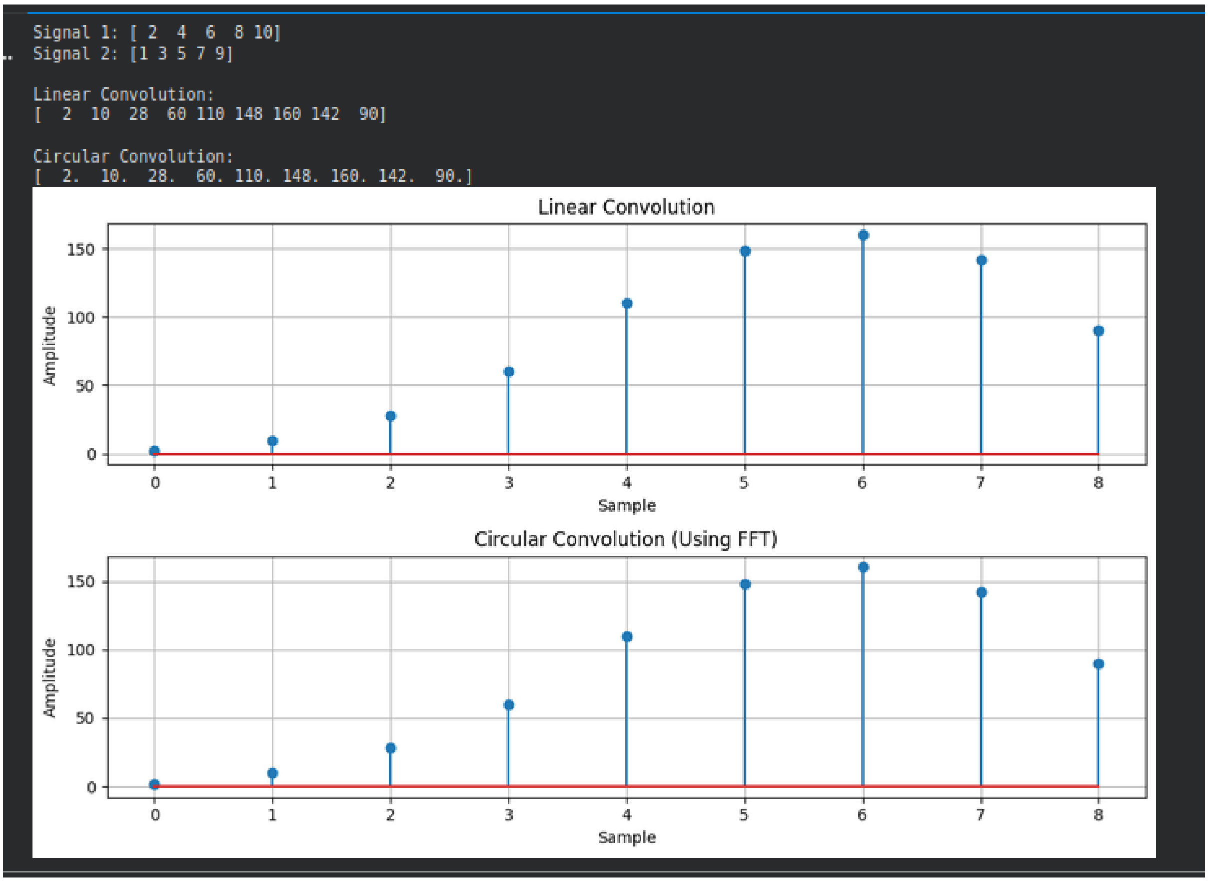
Code :

```
import numpy as np

import matplotlib.pyplot as plt

# Function to compute linear convolution

def linear_convolution(signal1, signal2):

    return np.convolve(signal1, signal2, mode='full')

# Function to compute circular convolution

def circular_convolution(signal1, signal2):

    fft_length = len(signal1) + len(signal2) - 1

    fft_signal1 = np.fft.fft(signal1, fft_length)

    fft_signal2 = np.fft.fft(signal2, fft_length)

    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)

    return circular_conv.real  # Take only the real part

# Define the discrete-time signals

signal1 = np.array([1, 2, 3, 4, 5])

signal2 = np.array([2, 4, 6, 8, 10])

# Compute the linear convolution

linear_conv = linear_convolution(signal1, signal2)

# Compute the circular convolution

circular_conv = circular_convolution(signal1, signal2)

 # Display the results

print("Signal 1:", signal1)
```

```
print("Signal 2:", signal2)

print("\nLinear Convolution:")

print(linear_conv)

print("\nCircular Convolution:")

print(circular_conv)

# Plot the results

plt.figure(figsize=(10, 6))

# Linear Convolution

plt.subplot(2, 1, 1)

plt.stem(range(len(linear_conv)), linear_conv)

plt.title("Linear Convolution")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

plt.grid(True)

# Circular Convolution

plt.subplot(2, 1, 2)

plt.stem(range(len(circular_conv)), circular_conv)

plt.title("Circular Convolution (Using FFT)")

plt.xlabel("Sample")

plt.ylabel("Amplitude")

plt.grid(True)

plt.tight_layout()

plt.show()

# Optional: Save the results
```

```
# np.savetxt("linear_convolution.txt", linear_conv, delimiter=",")

# np.savetxt("circular_convolution.txt", circular_conv, delimiter=",")
```

OUTPUT :



Conclusion : This experiment helped me understand the difference between linear convolution and circular convolution. Linear convolution gives a longer output, while circular convolution gives an output of the same length as the input signals. Using Python made it easy to perform the calculations and visualize the results.

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Experiment - 2 Calculation

Q → Linear Circular

signal 1 = [2, 4, 6, 8, 10]

Signal 2 = [1, 3, 5, 7, 9]

	1	3	5	7	9
2	2	6	10	14	18
4	4	12	20	28	36
6	6	18	30	42	54
8	8	24	40	56	72
10	10	30	50	70	90

$y(0) = 2$

$y(1) = 4 + 6 = 10$

$y(2) = 6 + 12 + 10 = 28$

$y(3) = 8 + 18 + 20 + 14 = 60$

$y(4) = 10 + 24 + 30 + 28 + 18 = 110$

$y(5) = 30 + 40 + 42 + 36 = 148$

$y(6) = 50 + 56 + 54 = 160$

$y(7) = 70 + 72 = 142$

$y(8) = 90$

$x(n) * h(n) = \{2, 10, 28, 60, 110, 148, 160, 142, 90\}$

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Circular Convolution

$$
\begin{bmatrix} 1 \\ 3 \\ 5 \\ 7 \\ 9 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot
\begin{bmatrix}
2 & 0 & 0 & 0 & 0 & 10 & 8 & 6 & 4 \\
4 & 2 & 0 & 0 & 0 & 0 & 10 & 8 & 6 \\
6 & 4 & \cancel{6}2 & 0 & 0 & 0 & 0 & 10 & 8 \\
8 & 6 & 4 & 2 & 0 & 0 & 0 & 0 & 10 \\
10 & 8 & 6 & 4 & 2 & 0 & 0 & 0 & 0 \\
0 & 10 & 8 & 6 & 4 & 2 & 0 & 0 & 0 \\
0 & 0 & 10 & 8 & 6 & 4 & 2 & 0 & 0 \\
0 & 0 & 0 & 10 & 8 & 6 & 4 & 2 & 0 \\
0 & 0 & 0 & 0 & 10 & 8 & 6 & 4 & 2
\end{bmatrix}
$$

$$
= \begin{bmatrix}
2 \\
4+6 \\
6+12+10 \\
8+16+20+14 \\
10+24+30+28+18 \\
30+40+42+36 \\
50+56+54 \\
70+72 \\
90
\end{bmatrix}
= \begin{bmatrix} 2 \\ 10 \\ 28 \\ 60 \\ 110 \\ 148 \\ 160 \\ 142 \\ 90 \end{bmatrix}
$$